

Anjali Krishnan A R

Software Engineer | C++ | Qt/QML | Systems Programming

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SUMMARY

Software Engineer with 1+ year of experience building **C++ and Qt/QML applications** on Linux and Windows. Hands-on experience developing forensic tools, system-level modules, and data processing pipelines at **C-DAC**, one of India's premier R&D organisations. Strong foundation in OOP, debugging, log analysis, and modular design. Seeking roles in C++ software development, embedded systems, or systems programming.

SKILLS

Languages	C, C++, Python, QML
Frameworks & UI	Qt5/Qt6, Qt Quick, Qt/QML
Core Concepts	OOP, Data Structures, Socket Programming, Multithreading, Modular Design
Tools	Git, CMake, Wireshark, Linux, Windows

WORK EXPERIENCE

Software Engineer

Dec 2024 – Present

Centre for Development of Advanced Computing (C-DAC), Thiruvananthapuram

- Designed and developed **C++ backend modules for IoT Forensics tools**, focusing on forensic data processing and cyber-attack detection in Smart Grid environments.
- Integrated backend forensic logic with **Qt/QML interfaces** for visualisation of evidence, timelines, and forensic reports.
- Processed system and application logs to support **event reconstruction and forensic investigations**.
- Performed testing, debugging, and validation of forensic software components to ensure accuracy and reliability.
- Prepared technical documentation and assisted in demonstrations of forensic tool functionalities.

Software Engineer Intern

Oct 2024 – Dec 2024

Quest Global Engineering Services

- Supported development and testing of C++ software components following object-oriented design principles.
- Assisted in debugging and validating system-level code, focusing on correctness and performance.

PROJECTS

1. IoT Forensics Tool for Smart Grid Systems | C++, Qt/QML, Windows

Developed a forensics software tool for Smart Grid environments to investigate cyber-attacks on connected IoT devices. Built data extraction and analysis modules to collect evidence from edge devices, gateways, and endpoints. Reconstructed event sequences and attack timelines from system logs following standard forensic procedures.

2. Network Packet Sniffer | C++, Linux, Raw Sockets

Built a command-line network packet capture tool using raw sockets in C++. Implemented protocol filtering (TCP/UDP/ICMP), real-time display of source/destination IPs and ports, and packet logging to file. Useful for network diagnostics and traffic analysis in forensic investigations.

3. TrueImager – Forensic Disk Imaging System | C++, Linux

Contributed to testing and validation of a forensic disk imaging tool. Verified disk acquisition, hashing, compression, and reporting features. Debugged device detection and imaging reliability issues across multiple storage devices.

4. Handwritten Text Recognition System | Python, Deep Learning, YOLO, TrOCR

Built a document processing pipeline combining YOLO for text region detection and TrOCR for handwritten text recognition. Implemented end-to-end image preprocessing, text extraction, and result generation with modular design.

EDUCATION

2020 – 2024 B.Tech in Computer Science and Engineering, **APJ Abdul Kalam Technological University**
GPA: 8.26/10.0